COMPANY CONFIDENTIAL

FSD Headquarters
Rockville, Maryland
November 21, 1960

Memorandum To: Mr. N. P. Edwards

SUBJECT: High Speed Computing Techniques Contract

At your suggestion, I have made a survey of the High Speed Computing Techniques contract -- its background and present status -- for the purpose of determining what went wrong last summer and of making recommendations as to what immediate action should be taken to assure its successful completion. During the survey, I have read all of the reports and correspondence available and have talked to most of the key people who have worked on the project both in Bethesda and Kingston. In addition, several discussions were held with Corporate Research personnel to get their opinions and informal recommendations.

Background

The High Speed Computing Techniques contract was begun in February, 1960, as the result of a successful proposal submitted in response to an RFP dated July 8, 1959. Rome Air Development Center was the administering agent for ARPA, who supplied the funds.

The work statement called for a study in such areas as: methods of high density storage, logical techniques for retrieval of solutions, numerical interpolation techniques, readout techniques to minimize memory access time, methods of pre-computing and recording solutions, and estimates of conventional equipment required to do the same tasks. The RFP did authorize the construction of bread-board experiments, but made no mention of problem analysis of applications per se.

Reports to be submitted included monthly progress letters, a technical note on June 30, 1960, and another at the end of the contract period (February, 1961).

The project was organized within IBM into three sections: a device survey task, exploratory design task, and problem analysis and techniques task. The first two were to be done at Kingston, the third at Bethesda.

Complications set in rather early when personnel on the project talked to both representatives of ARPA in Washington and with RADC personnel at Rome. This generated a "two boss" situation with IBM caught in between. As a result of ARPA guidance, the Bethesda group began an intensive study of the problem of multistatic radar detection of ballistic missiles and decoy discrimination. When Rome heard about this work later, they took serious exception to it on two grounds: (1) RADC was already funding studies in this area with Columbia University, and (2) The work statement said nothing about problem analysis.

This was briefly the situation at the time of the mid-term report. To add to the difficulties, there were internal disagreements between the Bethesda and Kingston versions of the mid-term report. As a result, there was a complete revision of the report written at the last moment (mostly by the Bethesda group). The report stressed the work which had been done on problem analysis since this did represent the bulk of the effort expended to date.

The mid-term report received a cool reception at Rome and several adverse statements were made concerning IBM's performance on the contract. (Incidentally, the recent stories of this type which have been heard are apparently just repetitions of these earlier statements and not new condemnations.)

In August, 1960, IBM reorganized the effort under Dr. R. J. Cypser and moved the whole project to Kingston. The NSRB agreed on October 17, 1960, to the writing of a continuation report at company expense. The report was written and submitted to Rome early in November. It is a well-written document and received favorable comment from all quarters.

A trip was made to Rome on November 16, 1960, by Dr. Cypser, Mr. Cox and others in which present status and plans were reviewed with RADC personnel. The general feeling was pleasant and indicated a willingness to forget the past and concentrate on achieving depth of coverage in the final report.

Causes of the Past Difficulties

In trying to assess the causes of the past difficulties, it is hard to point to any one serious mistake. The heavy emphasis on problem analysis, when none was called for in the work statement, can be pointed out. As a practical matter, some such analysis is absolutely necessary as background for the general computing techniques. Other shortcomings notable were: allowing the "two boss" situation to develop, following directions which were not given in writing, the lack of Marketing support which should have given warning and guidance,

the failure to make more use of Corporate Research talent, permitting the Bethesda-Kingston rivalry to build up, a general lack of interest and direction in the program (particularly at Kingston) before the reorganization. Certainly the government supervision of this contract has not been particularly helpful nor constructive. The planned changing of responsibility from one group in Rome to another has added to the confusion.

The fundamental cause of difficulty is probably due to the fact that this basically research-type contract was given to a development group to perform rather than to a research group.

Present Status

At the present time, each of the groups working on the project are making plans for their contributions to the final report. They are working hard and their morale seems good considering the criticism to which they have been subjected. They are "running scared".

There needs to be additional effort put in the area of mathematical techniques. Unfortunately some of this seems beyond the depth of the personnel presently assigned to the tasks. Additional support from Corporate Research or other experts in their fields is needed.

As an example: the work on the use of pre-computed solutions has languished because the examples considered have been radar-based problems which are characterized by large amounts of data and small amounts of computing. In such problems, pre-computed solutions save little if any time and require enormous memories. Instead, problems such as the solutions to partial differential equations which require little data but hours of computing should be examined to find real payoffs for this technique.

A seminar on intelligence applications is being planned for November 30 and December 1, at Bethesda. This should be quite helpful in discussing problems in the areas of information retrieval, indexing, etc. If handled properly, this can be a valuable input to the final report.

Recommendations for Immediate Action

1. The contract should continue under its present organization until the end of the present contract period (February, 1961). It is felt that another change of management or a redirection at this late date would insure failure, while the present efforts, provided corrective action is taken, are reasonably sure of success.

2. It is imperative that help be obtained from Corporate Research in certain technical areas. This help should be in the form of "term papers" on the subjects in question which can be incorporated into the final report with minor revisions. The general subject areas and some suggested individuals are:

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| a. Multivariate interpolation techniques | - L. H. Thomas<br>Frank Beckman |
| b. Problems in which pre-computed solutions would afford great time savings | - Ralph Willoughby<br>Dan Quarles<br>B. Gilchrist |
| c. High speed indexing techniques (Information retrieval) | - T. T. Tanimoto<br>A. Warheit |
| d. High speed statistical techniques | - Carl Kossack<br>M. Muller |

3. Help from Corporate Research and others should also be obtained in the form of an overall review of the contents of the final report as it is being pulled together. There is a great need for critical evaluation of the report to keep it short, clear, concise, and internally consistent.

I suggest that Dr. E. N. Adams of Research be asked to assist in this area.

4. Finally, it is recommended that any follow-on contract in this area be assigned to Corporate Research as their responsibility. This should be done with the understanding that several of the present FSD team members must be pledged to continue working in their areas of interest on the new contract to maintain a continuity of effort.

H. G. Kolsky
Senior Engineer

HGK/del

cc: Dr. E. N. Adams
Dr. R. J. Cypser